

Core Java (Questions 1–10)

1. Which component of the Java Architecture is responsible for converting bytecode into machine-specific code at runtime?

- A) JVM Compiler
- B) JIT Compiler
- C) Class Loader
- D) Garbage Collector
- **Answer: B**

2. Consider the following code snippet:

```
Java
int[] arr = {1, 2, 3, 4};
for(int x : arr) {
    x = x * 2;
}
System.out.println(arr[0]);
```

What is the output?

- A) 2
- B) 1
- C) 0
- D) Compilation Error
- **Answer: B**

3. Which method should be used to sort an array in parallel using the ForkJoin framework available in Java 8 and later?

- A) Arrays.sort()
- B) Arrays.parallelSort()
- C) Collections.sort()
- D) Arrays.concurrentSort()
- **Answer: B**

4. Which of the following statements is true about Git and GitHub?

- A) Git is a cloud-based hosting service; GitHub is a local version control tool.

- B) Git is a distributed version control system; GitHub is a web-based hosting service for Git repositories.
- C) They are identical tools developed by the same open-source community.
- D) Git cannot function without an active internet connection to GitHub.
- **Answer: B**

5. How can you enforce strict encapsulation on a class field?

- A) Declare the field as public and provide no methods.
- B) Declare the field as protected and provide public getters.
- C) Declare the field as private and provide public getters and setters.
- D) Declare the field as final without initializing it.
- **Answer: C**

6. What is the output of the following code?

Java

```
String s1 = new String("Java");
String s2 = "Java";
System.out.println(s1 == s2);
```

- A) true
- B) false
- C) null
- D) Compilation Error
- **Answer: B**

7. Which class should be preferred in a multi-threaded environment where string modifications are frequent?

- A) String
- B) StringBuilder
- C) StringBuffer
- D) StringJoiner
- **Answer: C**

8. What does the Optional class introduced in Java 8 aim to prevent?

- A) ClassCastException
- B) NullPointerException
- C) ArrayIndexOutOfBoundsException

- D) IllegalArgumentException
- **Answer: B**

9. Can an abstract class implement an interface without implementing its abstract methods?

- A) No, it must implement all methods immediately.
- B) Yes, it can pass the implementation responsibility to its concrete subclasses.
- C) Yes, but only if the interface methods are declared static.
- D) No, abstract classes cannot implement interfaces.
- **Answer: B**

10. How can a subclass access a hidden method or field of a superclass when a variable name conflict occurs?

- A) Using the this keyword
- B) Using the super keyword
- C) Using the parent keyword
- D) By casting to Object
- **Answer: B**

Object-Oriented Programming & Advanced Java (Questions 11–20)

11. Which statement is true regarding static methods inside an interface?

- A) They are automatically inherited by implementing classes.
- B) They can be invoked only by using the interface name.
- C) They must be declared with the public access modifier only.
- D) They can be overridden by the implementing subclass.
- **Answer: B**

12. What happens if you try to inherit from a class that is declared with the final keyword?

- A) The code compiles successfully but generates a runtime warning.
- B) A compilation error occurs.
- C) The subclass inherits only the public methods.
- D) The JVM bypasses the restriction automatically.
- **Answer: B**

13. Which keyword is used to explicitly import static members of a class into another file?

- A) import
- B) include static
- C) import static
- D) using static
- **Answer: C**

14. Which of the following exceptions is a checked exception in Java?

- A) NullPointerException
- B) IOException
- C) ArrayIndexOutOfBoundsException
- D) ArithmeticException
- **Answer: B**

15. What is the correct order of blocks when handling exceptions?

- A) try, finally, catch
- B) catch, try, finally
- C) try, catch, finally
- D) finally, try, catch
- **Answer: C**

16. What is the process of automatically converting a primitive type to its corresponding wrapper object called?

- A) Unboxing
- B) Autoboxing
- C) Upcasting
- D) Specialization
- **Answer: B**

17. Which interface in the Java Collection Framework does NOT inherit from the Collection interface?

- A) List
- B) Set
- C) Map
- D) Queue

- **Answer: C**

18. What is the primary benefit of using Generics in Java?

- A) Increasing program execution speed at runtime.
- B) Providing compile-time type safety and eliminating explicit typecasts.
- C) Reducing the size of the compiled bytecode file.
- D) Enabling automatic multithreading.
- **Answer: B**

19. Which intermediate operation is used in the Stream API to transform each element of a stream into another form?

- A) filter()
- B) map()
- C) forEach()
- D) collect()
- **Answer: B**

20. What is a functional interface?

- A) An interface with no methods.
- B) An interface with exactly one abstract method.
- C) An interface containing only default methods.
- D) An interface marked with the final keyword.
- **Answer: B**

Multithreading, Concurrency, and Design Patterns (Questions 21–30)

21. Which method must be implemented when a class implements the Runnable interface?

- A) start()
- B) execute()
- C) run()
- D) init()
- **Answer: C**

22. What occurs when two or more threads are blocked forever, each waiting for the resource held by the other?

- A) Starvation
- B) Deadlock
- C) Race Condition
- D) Livelock
- **Answer: B**

23. Which keyword ensures that changes made by one thread to a shared variable are immediately visible to all other threads?

- A) synchronized
- B) volatile
- C) final
- D) transient
- **Answer: B**

24. Which of the following classes from java.util.concurrent represents a thread-safe implementation of a hash map?

- A) Hashtable
- B) SynchronizedMap
- C) ConcurrentHashMap
- D) TreeMap
- **Answer: C**

25. Which design pattern ensures that a class has only one instance and provides a global point of access to it?

- A) Factory Pattern
- B) Observer Pattern
- C) Singleton Pattern
- D) Adapter Pattern
- **Answer: C**

26. In JUnit, which annotation is used to execute a method before each test case runs?

- A) @BeforeClass
- B) @BeforeEach
- C) @Test
- D) @SetUp
- **Answer: B**

27. Which JUnit assert method verifies that two objects point to the exact same memory location?

- A) assertEquals()
- B) assertTrue()
- C) assertSame()
- D) assertNotNull()
- **Answer: C**

28. What is a JUnit Test Suite used for?

- A) Aggregating and running multiple test classes together as a unit.
- B) Generating automatic test documentation.
- C) Mocking database calls.
- D) Stress testing application performance.
- **Answer: A**

29. Which pattern belongs to the behavioral category and defines a one-to-many dependency so that when one object changes state, all its dependents are notified?

- A) Factory
- B) Singleton
- C) Observer
- D) Builder
- **Answer: C**

30. What pattern should be used when the object creation logic needs to be hidden from the client, and objects are generated based on input arguments?

- A) Prototype Pattern
- B) Factory Method Pattern
- C) Proxy Pattern
- D) Facade Pattern
- **Answer: B**

Database, RDBMS, and JDBC (Questions 31–40)

31. Which of the following commands belongs to Data Definition Language (DDL)?

- A) INSERT
- B) UPDATE
- C) CREATE
- D) SELECT
- **Answer: C**

32. In a SQL statement, which clause is used to filter rows returned by a query based on a specific condition?

- A) ORDER BY
- B) GROUP BY
- C) WHERE
- D) HAVING
- **Answer: C**

33. Which JDBC interface is used to execute pre-compiled SQL queries with placeholder parameters?

- A) Statement
- B) PreparedStatement
- C) CallableStatement
- D) ResultSet
- **Answer: B**

34. What step must be performed first to open a connection using JDBC?

- A) Execute a query.
- B) Call DriverManager.getConnection().
- C) Instantiate a ResultSet object.
- D) Commit the transaction.
- **Answer: B**

35. How can you prevent an explicit transaction from saving changes permanently in case of an error?

- A) Call connection.commit()
- B) Call connection.rollback()
- C) Close the connection immediately.
- D) Throw a SQLException.
- **Answer: B**

36. Which object provides structural details about a database table or execution environment, such as column count and types?

- A) ResultSet
- B) DatabaseMetaData / ResultSetMetaData
- C) Statement
- D) Savepoint
- **Answer: B**

37. Which JDBC interface is explicitly designed to execute database stored procedures?

- A) PreparedStatement
- B) CallableStatement
- C) Driver
- D) RowSet
- **Answer: B**

38. What does a SELECT statement return?

- A) A single integer value representing affected rows.
- B) A structural schema definition.
- C) A tabular dataset known as a result set.
- D) An execution plan graph.
- **Answer: C**

39. Which SQL keyword is used to sort the output of a query in descending order?

- A) SORT BY
- B) ORDER BY ... DESC
- C) GROUP BY ... DESC
- D) ARRANGE
- **Answer: B**

40. What is the fundamental difference between DML and DDL commands?

- A) DML alters data values; DDL defines structural objects.
- B) DDL modifies row contents; DML sets user access rights.
- C) DML commands cannot be rolled back; DDL commands can be.
- D) There is no structural difference.

- **Answer: A**

Frontend Web Development: HTML, CSS, and Bootstrap (Questions 41–50)

41. Which HTML5 tag represents semantic structure indicating self-contained composition intended to be independently distributable?

- A) <div>
- B) <section>
- C) <article>
- D) <aside>
- **Answer: C**

42. What technology allows scalable vector graphics markup to be directly embedded inside HTML documents?

- A) JPEG
- B) SVG
- C) PNG
- D) GIF
- **Answer: B**

43. Which Web Storage mechanism allows data to persist even after the browser window or tab is closed?

- A) sessionStorage
- B) localStorage
- C) Cookie
- D) IndexedDB
- **Answer: B**

44. What does the CSS box model consist of, from the innermost component outward?

- A) Content, Margin, Border, Padding
- B) Content, Padding, Border, Margin
- C) Content, Border, Padding, Margin
- D) Margin, Border, Padding, Content
- **Answer: B**

45. Which of the following is an example of a CSS Preprocessor?

- A) Bootstrap
- B) SASS
- C) Tailwind
- D) jQuery
- **Answer: B**

46. What CSS Grid property defines the space or gutter gap between grid tracks?

- A) grid-template-columns
- B) grid-gap (or gap)
- C) margin
- D) padding
- **Answer: B**

47. How many default grid column tracks form a row in the standard Bootstrap layout system?

- A) 6
- B) 10
- C) 12
- D) 16
- **Answer: C**

48. Which class in Bootstrap is utilized to create a responsive, collapsing navigation header bar?

- A) .navbar
- B) .nav-tabs
- C) .menu-bar
- D) .header-nav
- **Answer: A**

49. What CSS property is essential to initiate a smooth property transition or keyframe animation?

- A) display
- B) position
- C) transition / animation
- D) float
- **Answer: C**

50. What design practice focuses on making websites usable for all individuals, including those with visual, auditory, or motor impairments?

- A) SEO Optimization
- B) Typography
- C) Accessibility (A11y)
- D) Localization
- **Answer: C**

JavaScript & Client-Side Scripting (Questions 51–60)

51. Which keyword should be avoided in modern ES6+ JavaScript to prevent scope leakage out of code blocks?

- A) let
- B) const
- C) var
- D) function
- **Answer: C**

52. What mechanism allows an inner JavaScript function access to the outer enclosing function's variables even after the outer function finishes executing?

- A) Prototype Chain
- B) Closure
- C) Hoisting
- D) Event Bubbling
- **Answer: B**

53. Which JavaScript object represents an asynchronous operation that may eventually resolve with a value or reject with an error?

- A) JSON
- B) Promise
- C) AJAX
- D) Deferred
- **Answer: B**

54. What standard JavaScript object syntax format is widely utilized for client-server text data interchange?

- A) XML
- B) YAML
- C) JSON
- D) INI
- **Answer: C**

55. What method is used to parse a JSON-formatted string into a native JavaScript object?

- A) JSON.stringify()
- B) JSON.parse()
- C) JSON.objectify()
- D) JSON.toObject()
- **Answer: B**

56. What pattern delegates a single event handler to a common parent element rather than attaching event listeners to individual child elements?

- A) Event Capturing
- B) Event Delegation
- C) Event Propagation
- D) Event Throttling
- **Answer: B**

57. Which modern native API replaces old XMLHttpRequest setups to fetch remote network assets using promises?

- A) ajax()
- B) fetch()
- C) axios()
- D) http()
- **Answer: B**

58. What is the main objective of optimizing client-side performance through code minification and DOM reduction?

- A) Making the codebase more legible.
- B) Speeding up page rendering and interactive response metrics.
- C) Bypassing modern browser firewall restrictions.
- D) Encrypting client-side JavaScript.
- **Answer: B**

59. What does the Document Object Model (DOM) represent?

- A) The underlying structural database records layout.
- B) A structural object representation of the rendered HTML content node tree.
- C) An encryption key wrapper.
- D) The styles file stylesheet layout.
- **Answer: B**

60. How do you declare a block-scoped block immutable variable constant in ES6?

- A) let
- B) var
- C) const
- D) static
- **Answer: C**

Build Tools & Fundamentals of Web Applications: Maven, Servlets, and JSP (Questions 61–70)

61. What XML configuration file defines a project's metadata, plugins, and dependencies in an Apache Maven setup?

- A) settings.xml
- B) pom.xml
- C) build.xml
- D) manifest.mf
- **Answer: B**

62. Which command represents the standard Maven build lifecycle phase that compiles code, runs unit tests, and packages it into a JAR/WAR file?

- A) mvn clean
- B) mvn compile
- C) mvn package
- D) mvn archetype:generate
- **Answer: C**

63. What is Maven Archetype used for?

- A) Managing remote artifact servers.
- B) Generating an initial project boilerplate template structure.
- C) Analyzing source code errors.
- D) Handling test mocking metrics.
- **Answer: B**

64. Which class serves as the core base class for creating components that handle HTTP requests and generate dynamic responses in Java web applications?

- A) GenericServlet
- B) HttpServlet
- C) FacesServlet
- D) Filter
- **Answer: B**

65. Which method inside a custom servlet handles incoming HTTP POST requests?

- A) doGet()
- B) doPost()
- C) service()
- D) init()
- **Answer: B**

66. What technology enables developers to interleave HTML code with server-side Java expressions using expression tags like `<%= %>`?

- A) HTMLTemplates
- B) JavaServer Pages (JSP)
- C) JavaScript
- D) CSS-in-JS
- **Answer: B**

67. Which standard action tag is used in JSP to locate, bind, or instantiate a reusable data object class?

- A) `<jsp:useBean>`
- B) `<jsp:setProperty>`
- C) `<jsp:forward>`
- D) `<jsp:include>`

- **Answer: A**

68. What phase of the Maven lifecycle removes the target compilation output folder to prepare for a fresh rebuild?

- A) install
- B) clean
- C) validate
- D) deploy
- **Answer: B**

69. Where are external binary JAR files declared to be automatically resolved from remote repositories by Maven?

- A) <plugins> block
- B) <dependencies> block
- C) <build> block
- D) <properties> block
- **Answer: B**

70. What is the explicit scope of a Java web container component instance life cycle?

- A) Controlled via JVM Garbage Collection rules alone.
- B) Controlled entirely by the container's specialized methods (init(), service(), destroy()).
- C) Determined by database connections open states.
- D) Created statically at operating system boot runtime.
- **Answer: B**

Spring Framework Core & Dependency Injection (Questions 71–80)

71. What core programming paradigm underpins the Spring Framework, shifting component instantiation control from application code to the framework?

- A) Aspect-Oriented Programming (AOP)
- B) Inversion of Control (IoC) / Dependency Injection (DI)
- C) Functional Reactive Programming
- D) Model-View-Controller
- **Answer: B**

72. Which container interface represents the advanced form of Spring IoC container, adding enterprise integration features over standard BeanFactory?

- A) BeanFactory
- B) ApplicationContext
- C) BeanRegistry
- D) ServletConfig
- **Answer: B**

73. What is the default lifecycle scope of a bean managed within a Spring framework context?

- A) Prototype
- B) Singleton
- C) Request
- D) Session
- **Answer: B**

74. Which type of Dependency Injection injects components using class constructor parameters?

- A) Setter Injection
- B) Interface Injection
- C) Constructor Injection
- D) Field Injection
- **Answer: C**

75. Which annotation instructs Spring to inject a matching managed bean dependency automatically?

- A) @Component
- B) @Service
- C) @Autowired
- D) @Bean
- **Answer: C**

76. When multiple beans of the exact same type exist, which annotation can resolve ambiguity by specifying a unique bean name?

- A) @Primary
- B) @Qualifier

- C) @Resource
- D) @Component
- **Answer: B**

77. Which configuration style uses Java annotations like @Configuration and @Bean instead of XML files to configure Spring contexts?

- A) XML Configuration
- B) Java-Based Configuration
- C) Properties Configuration
- D) Manifest Configuration
- **Answer: B**

78. What happens to a bean declared with the prototype scope each time it is requested from the Spring container?

- A) The same cached instance is returned.
- B) A new, distinct bean instance is created and returned.
- C) An exception is raised.
- D) The container resets entirely.
- **Answer: B**

79. What annotation is applied at the class level to register an ordinary class as a managed bean component within the Spring application context?

- A) @Bean
- B) @Component
- C) @Inject
- D) @Scope
- **Answer: B**

80. Which component lifecycle phase runs immediately after bean properties have been initialized by the IoC container?

- A) @PostConstruct (or custom initialization method)
- B) @PreDestroy
- C) Constructor execution
- D) Finalize phase
- **Answer: A**

Spring MVC & Hibernate ORM (Questions 81–90)

81. Which core component acts as the Front Controller inside a Spring MVC web application?

- A) HandlerMapping
- B) ViewResolver
- C) DispatcherServlet
- D) ModelAndView
- **Answer: C**

82. Which Spring MVC component determines exactly which Controller class handles an incoming URL request?

- A) ViewResolver
- B) HandlerMapping
- C) DispatcherServlet
- D) ModelMap
- **Answer: B**

83. What object translates abstract logical view names returned from controllers into physical rendering view resources (like JSP or HTML pages)?

- A) DispatcherServlet
- B) ViewResolver
- C) HandlerAdapter
- D) ModelAndView
- **Answer: B**

84. What persistent layer issue does Object-Relational Mapping (ORM) solve?

- A) Network routing speed mismatch.
- B) The structural paradigm mismatch between object-oriented programming models and relational tables layout.
- C) Cross-browser layout display inconsistencies.
- D) Thread deadlock runtime errors.
- **Answer: B**

85. Which object acts as a factory for acquiring database connection sessions within standard Hibernate configurations?

- A) Configuration
- B) SessionFactory

- C) Session
- D) Transaction
- **Answer: B**

86. Which Hibernate object state describes an entity instance that has an associated identifier but is no longer actively managed by a running persistent session?

- A) Transient
- B) Persistent
- C) Detached
- D) Removed
- **Answer: C**

87. What high-level object-oriented query language does Hibernate provide to query database entities instead of raw tables?

- A) SQL
- B) HQL (Hibernate Query Language)
- C) JPQL
- D) XML-QL
- **Answer: B**

88. Which API allows queries to be constructed programmatically through strongly typed object criteria elements rather than text strings?

- A) HQL API
- B) Criteria API
- C) Native SQL API
- D) JDBC API
- **Answer: B**

89. Which annotation designates a standard Java POJO class as a persistent entity mapped to a database table?

- A) @Table
- B) @Entity
- C) @Id
- D) @Persistent
- **Answer: B**

90. What annotation defines the primary key identifier attribute of a Hibernate entity?

- A) @Column
- B) @GeneratedValue
- C) @Id
- D) @Key
- **Answer: C**

Spring Boot, Spring Security & Enterprise REST APIs (Questions 91–100)

91. What mechanism allows Spring Boot to configure application components automatically based on the JAR libraries present in the classpath?

- A) Auto-Configuration
- B) Starters
- C) Actuator
- D) DevTools
- **Answer: A**

92. Which module provides enterprise-ready monitoring features like health checks, metrics, and thread dumps out-of-the-box in Spring Boot?

- A) DevTools
- B) Actuator
- C) Security
- D) Thymeleaf
- **Answer: B**

93. What annotation combines @Controller and @ResponseBody to create a dedicated web endpoint optimized for RESTful JSON web service responses?

- A) @RestController
- B) @Service
- C) @Repository
- D) @Endpoint
- **Answer: A**

94. What template engine is natively integrated as standard starter views for server-side HTML UI rendering in modern Spring Boot apps?

- A) JSP
- B) Thymeleaf
- C) Angular
- D) JSF
- **Answer: B**

95. Which Spring Boot module automatically restarts the local embedded server whenever project code files undergo recompilation changes?

- A) Actuator
- B) DevTools
- C) Starter Parent
- D) Initializr
- **Answer: B**

96. What token mechanism allows secure, stateless representation of user identity data to be passed across distributed enterprise microservices?

- A) Basic Auth Token
- B) Session Cookies
- C) JWT (JSON Web Token)
- D) JID Token
- **Answer: C**

97. Which template class reduces boilerplate code when interacting with traditional relational databases using standard SQL via JDBC in Spring?

- A) JdbcTemplate
- B) SessionTemplate
- C) JpaRepository
- D) TransactionTemplate
- **Answer: A**

98. Which annotation can handle mapping variations between complex database entity tables relationships, like a parent holding multiple children?

- A) @Column
- B) @OneToMany
- C) @Transient
- D) @Embedded
- **Answer: B**

99. What database type represents an embedded in-memory database used for rapid local unit testing execution setups?

- A) Oracle DB
- B) MongoDB
- C) H2 Database
- D) PostgreSQL
- **Answer: C**

100. Which tool generates automated, dynamic documentation layouts and execution environments for exposed REST endpoints?

- A) JUnit
- B) Swagger (OpenAPI)
- C) Mockito
- D) Zipkin
- **Answer: B**

Microservices Architecture, Cloud Native, and Security (Questions 101–110)

101. Which architectural style replaces single monolithic applications with collections of small, loosely coupled, independently deployable software services?

- A) Monolithic Architecture
- B) Microservices Architecture
- C) Client-Server Architecture
- D) Mainframe Architecture
- **Answer: B**

102. What component acts as a service discovery registry server where microservices register themselves so others can locate them?

- A) Spring Cloud LoadBalancer
- B) Spring Cloud Gateway
- C) Netflix Eureka Server
- D) Zipkin Server
- **Answer: C**

103. Which declarative REST client tool simplifies HTTP web service calls between microservices by using interface annotations?

- A) RestTemplate
- B) OpenFeign
- C) WebClient
- D) HttpClient
- **Answer: B**

104. What design pattern prevents cascading service failures across an enterprise microservice graph by returning alternative responses when a target fails?

- A) Service Discovery Pattern
- B) Circuit Breaker Pattern
- C) API Gateway Pattern
- D) CQRS Pattern
- **Answer: B**

105. Which library serves as the primary modern solution for implementing the Circuit Breaker pattern within Spring Cloud ecosystems?

- A) Hystrix
- B) Resilience4J
- C) Ribbon
- D) Zuul
- **Answer: B**

106. What centralized tool is used to monitor circuit breaker states and aggregate performance metrics in cloud deployments?

- A) Prometheus Dashboard
- B) Config Server
- C) Eureka
- D) Sleuth
- **Answer: A**

107. Which component provides unified configuration management across all microservices environments through an externalized git backend repository?

- A) Spring Cloud Configuration Server
- B) API Gateway
- C) Zipkin Tracker
- D) Maven Central

- **Answer: A**

108. What architectural component acts as a single point of entry to route, filter, and secure all incoming client requests to backend microservices?

- A) Eureka Registry
- B) Spring Cloud Gateway
- C) Config Server
- D) Circuit Breaker
- **Answer: B**

109. Which tool combination enables distributed log tracing by injecting unique trace IDs into asynchronous headers across microservices?

- A) Spring Cloud Sleuth and Zipkin
- B) Prometheus and Grafana
- C) Eureka and Ribbon
- D) Maven and Jenkins
- **Answer: A**

110. What security mechanism allows restricted web resources on a server to be requested by a web application originating from a different domain?

- A) Basic Auth
- B) JWT Token
- C) CORS (Cross-Origin Resource Sharing)
- D) Ant Matchers
- **Answer: C**

Advanced Java Mechanics & Best Practices (Questions 111–120)

111. Which block in Java is guaranteed to execute regardless of whether an exception is thrown or caught?

- A) catch
- B) try
- C) finally
- D) throw
- **Answer: C**

112. What mechanism is used by Java's Garbage Collector to identify objects that are candidates for deletion?

- A) Reference Counting only
- B) Root Reachability Analysis
- C) Manual memory tracking
- D) Method size monitoring
- **Answer: B**

113. What is the main advantage of StringJoiner over manual string concatenation in loops?

- A) It makes strings mutable permanently.
- B) It optimizes delimiter, prefix, and suffix formatting with high efficiency.
- C) It runs on a separate background thread.
- D) It avoids the creation of class definitions.
- **Answer: B**

114. Which of the following is true about a Java package?

- A) It prevents classes from being compiled.
- B) It provides a namespace partition to avoid naming collisions and structure code.
- C) It speeds up database access.
- D) It enforces singleton patterns automatically.
- **Answer: B**

115. What keyword is used to pass an exception up the call stack from a method signature?

- A) throw
- B) throws
- C) try
- D) catch
- **Answer: B**

116. Which collector method should be used to transform a Stream into an unmodifiable List?

- A) Collectors.toList()
- B) Collectors.toUnmodifiableList()

- C) Collectors.toSet()
- D) Collectors.toMap()
- **Answer: B**

117. What interface represents a function that accepts one argument and returns a boolean value?

- A) Function
- B) Consumer
- C) Predicate
- D) Supplier
- **Answer: C**

118. What lambda symbol separates parameters from the execution body?

- A) ::
- B) ->
- C) =>
- D) .
- **Answer: B**

119. What shorthand syntax allows referencing methods directly using their class or instance name without invoking them?

- A) Lambda Expressions
- B) Method References (::)
- C) Closures
- D) Anonymous classes
- **Answer: B**

120. Which exception is thrown if you attempt to modify structurally an unmodifiable list collection?

- A) NullPointerException
- B) UnsupportedOperationException
- C) ConcurrentModificationException
- D) IllegalArgumentException
- **Answer: B**

Multi-Threading & Concurrency Deep Dive (Questions 121–130)

121. What happens if a thread calls join() on another thread?

- A) The current thread stops execution permanently.
- B) The current thread pauses until the targeted thread finishes execution.
- C) The two threads are merged into a single thread structure.
- D) A deadlock condition is immediately created.
- **Answer: B**

122. Which framework provides thread pools to manage and schedule worker threads efficiently in concurrent Java applications?

- A) Stream API
- B) Executor Framework
- C) Collections Framework
- D) Serialization Engine
- **Answer: B**

123. What lock primitive provides more flexible, fine-grained locking options than synchronized blocks?

- A) volatile
- B) ReentrantLock
- C) AtomicInteger
- D) ThreadLocal
- **Answer: B**

124. Which data structure is explicitly optimized for read-heavy operations where write modifications are infrequent?

- A) ArrayList
- B) CopyOnWriteArrayList
- C) Vector
- D) LinkedList
- **Answer: B**

125. What atomic class guarantees lock-free, thread-safe updates on an integer variable?

- A) Integer
- B) VolatileInteger
- C) AtomicInteger

- D) SynchronizedInteger
- **Answer: C**

126. What does thread synchronization achieve?

- A) Faster network I/O execution.
- B) Preventing concurrent race conditions by restricting access to a shared resource to one thread at a time.
- C) Automatic code minification.
- D) Eliminating the need for heap memory allocation.
- **Answer: B**

127. Which method yields the current thread's execution lease to allow other equal-priority threads an opportunity to run?

- A) Thread.sleep()
- B) Thread.yield()
- C) Thread.stop()
- D) Object.wait()
- **Answer: B**

128. What is a key practice for ensuring complete thread safety?

- A) Keeping objects mutable and globally accessible.
- B) Minimizing shared mutable state and leveraging immutability wherever possible.
- C) Removing all final keywords from variable declarations.
- D) Running all code blocks inside a single global thread instance.
- **Answer: B**

129. Which class provides a synchronization barrier that allows a set of threads to wait for each other to reach a common barrier point?

- A) Semaphore
- B) CyclicBarrier
- C) CountdownLatch
- D) Exchanger
- **Answer: B**

130. What concurrency utility operates like a countdown counter to block threads until a set of operations completes?

- A) CountdownLatch
- B) CyclicBarrier
- C) Phaser
- D) ForkJoinPool
- **Answer: A**

Database Interaction, Filters & Advanced Queries (Questions 131–140)

131. Which clause is used to eliminate duplicate rows from a SQL SELECT statement output?

- A) UNIQUE
- B) DISTINCT
- C) SINGLE
- D) GROUP
- **Answer: B**

132. What is the purpose of the SQL LIKE operator?

- A) Comparing numeric values for equality.
- B) Performing pattern matching using wildcards inside text columns.
- C) Joining two independent database tables.
- D) Sorting data records in ascending order.
- **Answer: B**

133. What is a foreign key?

- A) A key that guarantees unique rows within its own native table.
- B) A column that establishes a link between data in two tables by referencing a primary key in another table.
- C) An unindexed placeholder field.
- D) A key generated exclusively for cloud backup tracking.
- **Answer: B**

134. Which JDBC method executes SQL statements that update, insert, or delete data records?

- A) executeQuery()
- B) executeUpdate()
- C) executeModify()
- D) executeAlter()

- **Answer: B**

135. What property of transactions guarantees that either all operations complete successfully or none are applied?

- A) Atomicity
- B) Consistency
- C) Isolation
- D) Durability
- **Answer: A**

136. What error occurs when a Java program attempts to execute an invalid SQL statement through JDBC?

- A) NullPointerException
- B) SQLException
- C) IOException
- D) ClassNotFoundException
- **Answer: B**

137. Which database constraint ensures that a column cannot accept NULL values?

- A) PRIMARY KEY
- B) NOT NULL
- C) UNIQUE
- D) CHECK
- **Answer: B**

138. What type of join returns all matching records from both tables, plus unmatched rows from the left table?

- A) Inner Join
- B) Left Outer Join
- C) Right Outer Join
- D) Full Outer Join
- **Answer: B**

139. Which keyword is combined with GROUP BY to filter aggregate groups, serving a similar purpose to the WHERE clause?

- A) HAVING
- B) WHERE AGGREGATE
- C) FILTER
- D) RESTRICT
- **Answer: A**

140. What is the default port commonly used by an Oracle Database instance?

- A) 3306
- B) 5432
- C) 1521
- D) 8080
- **Answer: C**

UI Engineering: Semantic Markup, CSS Grid & Preprocessors (Questions 141–150)

141. What HTML5 element is used to display independent, standalone auxiliary content like a sidebar alongside main articles?

- A) <section>
- B) <nav>
- C) <aside>
- D) <header>
- **Answer: C**

142. What attribute provides alternative text descriptions for images to enhance web accessibility?

- A) title
- B) alt
- C) src
- D) longdesc
- **Answer: B**

143. Which CSS rule applies specific styles only when the viewport matches defined criteria, such as a maximum width for mobile screens?

- A) @import
- B) @media
- C) @keyframes

- D) @font-face
- **Answer: B**

144. What layout paradigm allows alignment of items in a dual-axis grid pattern (columns and rows simultaneously)?

- A) Flexbox
- B) CSS Grid
- C) Floats
- D) Inline Blocks
- **Answer: B**

145. In SASS, what feature allows you to define reusable chunks of styles that can be included across multiple selectors?

- A) Variables
- B) Mixins
- C) Nesting
- D) Partials
- **Answer: B**

146. What framework practice builds styling architectures entirely inside JavaScript files alongside UI component logic?

- A) External CSS
- B) SASS Preprocessing
- C) CSS-in-JS
- D) Semantic HTML
- **Answer: C**

147. Which CSS property changes the layout flow behavior of an element to either inline, block, flex, or grid?

- A) position
- B) display
- C) overflow
- D) visibility
- **Answer: B**

148. What HTML5 attribute enables client-side storage of arbitrary custom data directly on HTML elements?

- A) class
- B) id
- C) data-*
- D) value
- **Answer: C**

149. What concept involves designing a web experience starting from the smallest screens and scaling up to desktops?

- A) Desktop-First Design
- B) Mobile-First Design
- C) Graceful Degradation
- D) Server-Side Optimization
- **Answer: B**

150. Which HTML5 tag is used to embed audio files natively without requiring external plug-ins?

- A) <media>
- B) <audio>
- C) <sound>
- D) <object>
- **Answer: B**

Advanced Client Scripting & Async JavaScript (Questions 151–160)

151. What execution status results from a JavaScript function that uses the async keyword?

- A) It returns a string.
- B) It always returns a Promise.
- C) It executes synchronously.
- D) It halts browser rendering.
- **Answer: B**

152. Which keyword is used within an async function to pause execution until a Promise resolves or rejects?

- A) wait
- B) defer
- C) await

- D) pause
- **Answer: C**

153. What issue does JavaScript variable hoisting present when variables are declared using the var keyword?

- A) The variable throws an error immediately.
- B) The variable declaration is moved to the top of its scope, making it accessible as undefined before initialization.
- C) The variable cannot be reassigned.
- D) The variable is deleted automatically.
- **Answer: B**

154. What block handles exceptions and run-time errors cleanly within modern asynchronous async/await code structures?

- A) .then().catch()
- B) try...catch
- C) if...else
- D) throw...finally
- **Answer: B**

155. Which array method creates a new array containing only elements that pass a specific condition?

- A) map()
- B) filter()
- C) reduce()
- D) forEach()
- **Answer: B**

156. What does DOM manipulation refer to in client-side web programming?

- A) Changing server-side database records.
- B) Programmatically altering the structure, style, or content of a web page using JavaScript.
- C) Minifying CSS source code files.
- D) Verifying secure network communication certificates.
- **Answer: B**

157. What concept prevents a function from being called too frequently by delaying its execution until a certain amount of time has elapsed?

- A) Event Delegation
- B) Debouncing
- C) Event Bubbling
- D) Hoisting
- **Answer: B**

158. What format is native to JavaScript and represents data as structured key-value pairs?

- A) XML Document
- B) JavaScript Object / JSON
- C) Markdown Text
- D) Bytecode array
- **Answer: B**

159. What does jQuery's AJAX method return to track the success or failure of asynchronous network requests?

- A) A raw string array
- B) A Deferred/Promise object
- C) A DOM node element
- D) A blank interface pointer
- **Answer: B**

160. What scope level is introduced when variables are declared using let or const inside a block?

- A) Global Scope
- B) Function Scope
- C) Block Scope
- D) Module Scope
- **Answer: C**

Spring Beans, Dependency Inversion, and Lifecycle (Questions 161–170)

161. What does Dependency Inversion mean in application architecture?

- A) Lower-level modules must drive higher-level logic directly.

- B) High-level modules should not depend on low-level modules; both should depend on abstractions.
- C) Database systems should generate application source code.
- D) Web browsers should handle application caching entirely.
- **Answer: B**

162. Which type of injection is ideal for enforcing mandatory dependencies that a class requires to function properly?

- A) Setter Injection
- B) Constructor Injection
- C) Field Injection
- D) Interface Injection
- **Answer: B**

163. What type of autowiring tells Spring to look for a bean with the exact same name as the property it needs to inject?

- A) byType
- B) byName
- C) constructor
- D) no
- **Answer: B**

164. What type of autowiring matches and injects dependencies based on their class type or interface?

- A) byName
- B) byType
- C) constructor
- D) explicit
- **Answer: B**

165. What annotation marks a class as a configuration source containing factory methods that define Spring beans?

- A) @Component
- B) @Configuration
- C) @Service
- D) @Controller
- **Answer: B**

166. Which annotation specifies that a bean should be initialized lazily rather than at startup?

- A) @Lazy
- B) @Scope("prototype")
- C) @Eager(false)
- D) @Delay
- **Answer: A**

167. What interface can a Spring bean implement if it needs to perform cleanup operations when the container is destroyed?

- A) InitializingBean
- B) DisposableBean
- C) Runnable
- D) Cloneable
- **Answer: B**

168. Which annotation can define a factory method inside a @Configuration class to register the returned object as a Spring bean?

- A) @Component
- B) @Bean
- C) @Autowired
- D) @Resource
- **Answer: B**

169. What mechanism allows Spring to scan the classpath and automatically detect classes annotated with stereotype annotations?

- A) Manual Registration
- B) Component Scanning
- C) XML Mapping
- D) Reflection Compilation
- **Answer: B**

170. Which stereotype annotation specifically designates a bean as part of the data access persistence layer?

- A) @Service
- B) @Repository

- C) @Controller
- D) @Component
- **Answer: B**

Advanced Web MVC Flow & Hibernate State Management (Questions 171–180)

171. What is the correct execution order of components in a Spring MVC request-response lifecycle?

- A) Client → Controller → DispatcherServlet → View
- B) Client → DispatcherServlet → HandlerMapping → Controller → ViewResolver → View
- C) Client → ViewResolver → Controller → DispatcherServlet
- D) Controller → HandlerMapping → DispatcherServlet → View
- **Answer: B**

172. Which annotation binds an incoming HTTP request parameter directly to a method argument in a Spring MVC controller?

- A) @PathVariable
- B) @RequestParam
- C) @RequestBody
- D) @ModelAttribute
- **Answer: B**

173. Which annotation extracts variables from the URI path segment itself (e.g., /users/{id})?

- A) @RequestParam
- B) @PathVariable
- C) @CookieValue
- D) @RequestHeader
- **Answer: B**

174. What is the role of form validation annotations like @NotNull or @Size in Spring MVC?

- A) Restricting database indexing speeds.
- B) Enforcing data integrity rules on domain model inputs before processing them.

- C) Automatically converting JSON strings into HTML files.
- D) Managing session expiration times.
- **Answer: B**

175. Which state applies to a newly instantiated Hibernate object that has no database identity and is not yet associated with an active session?

- A) Persistent
- B) Transient
- C) Detached
- D) Removed
- **Answer: B**

176. Which method saves an entity instance to the database and transitions its state to persistent?

- A) session.save() or session.persist()
- B) session.detach()
- C) session.clear()
- D) session.evict()
- **Answer: A**

177. What happens to changes made to a persistent entity during an active Hibernate transaction?

- A) They are ignored until the server restarts.
- B) They are tracked automatically and synchronized with the database when the transaction commits.
- C) They throw a compile-time error.
- D) They require manual execution of raw SQL update queries.
- **Answer: B**

178. Which method transitions a persistent entity into a detached state, removing it from session management?

- A) session.evict() or session.clear()
- B) session.save()
- C) session.update()
- D) session.flush()
- **Answer: A**

179. What does a call to session.flush() do in Hibernate?

- A) Commits the transaction and closes the database connection permanently.
- B) Forces Hibernate to synchronize its in-memory entity states with the underlying database immediately.
- C) Deletes all data from the database.
- D) Clears the second-level cache.
- **Answer: B**

180. What interface handles transactional boundaries programmatically or declaratively in Spring environments?

- A) PlatformTransactionManager
- B) Session
- C) Connection
- D) EntityManager
- **Answer: A**

Spring Boot Starters, Actuators & Security Configurations (Questions 181–190)

181. What prefix distinguishes standard Spring Boot starter dependency artifacts?

- A) spring-boot-starter-*
- B) spring-framework-starter-*
- C) boot-dependency-*
- D) starter-pack-*
- **Answer: A**

182. Which configuration file can customize application properties like server port and database URLs in a Spring Boot project?

- A) pom.xml
- B) application.properties (or application.yml)
- C) web.xml
- D) beans.xml
- **Answer: B**

183. Which annotation marks the main class of a Spring Boot application, enabling auto-configuration and component scanning?

- A) @SpringBootApplication
- B) @EnableAutoConfiguration
- C) @ComponentScan
- D) @Configuration
- **Answer: A**

184. What is the default embedded web container provided out-of-the-box by the spring-boot-starter-web module?

- A) Jetty
- B) Tomcat
- C) Undertow
- D) GlassFish
- **Answer: B**

185. Which filter chain handles security enforcement across incoming request patterns in Spring Security configurations?

- A) ServletFilterChain
- B) SecurityFilterChain
- C) InterceptorRegistry
- D) CorsFilter
- **Answer: B**

186. Which method or expression configures access control rules based on URL path patterns in Spring Security?

- A) antMatchers() (or requestMatchers())
- B) formLogin()
- C) httpBasic()
- D) csrf()
- **Answer: A**

187. What security standard transmits authentication credentials as a base64-encoded username and password pair within the HTTP Authorization header?

- A) OAuth2
- B) JWT Authentication
- C) Basic Authentication
- D) Digest Authentication

- **Answer: C**

188. Which component is responsible for loading user-specific data during authentication processes in Spring Security?

- A) AuthenticationManager
- B) UserDetailsService
- C) PasswordEncoder
- D) SecurityContextHolder
- **Answer: B**

189. What library encodes raw user passwords securely using a slow, cryptographic hashing function before saving them to a database?

- A) MD5
- B) SHA-256
- C) BCryptPasswordEncoder
- D) Base64Encoder
- **Answer: C**

190. What strategy pattern handles cross-cutting REST endpoint exceptions uniformly across an entire Spring application?

- A) try...catch blocks inside every controller method
- B) @ControllerAdvice combined with @ExceptionHandler methods
- C) Custom servlet filter declarations
- D) Database trigger configurations
- **Answer: B**

Cloud Resilience, Circuit Breakers & Distributed Tracing (Questions 191–200)

191. What cloud configuration pattern allows properties to be reloaded dynamically at runtime without restarting microservices?

- A) Dynamic Discovery
- B) @RefreshScope combined with Spring Cloud Config Server
- C) Recompiling build artifacts
- D) Re-executing Maven clean commands
- **Answer: B**

192. Which state indicates that a Circuit Breaker is routing all requests to the target service because it is functioning normally?

- A) OPEN
- B) HALF_OPEN
- C) CLOSED
- D) DISABLED
- **Answer: C**

193. What state does a Circuit Breaker transition to when its failure rate exceeds a predefined threshold, blocking all requests immediately?

- A) CLOSED
- B) OPEN
- C) HALF_OPEN
- D) ISOLATED
- **Answer: B**

194. Why does a Circuit Breaker periodically enter a HALF_OPEN state after a configured delay?

- A) To permanently shut down the system.
- B) To allow a limited number of trial requests through to check if the underlying service has recovered.
- C) To reset all database connections.
- D) To force a hard reboot of the cloud server cluster.
- **Answer: B**

195. What does Client-Side Load Balancing mean in a microservices ecosystem?

- A) The client browser handles load balancing between different applications.
- B) A calling microservice retrieves instances from a service registry and balances requests across them itself.
- C) A centralized hardware gateway routes all incoming traffic.
- D) Database servers balance their transactional workloads automatically.
- **Answer: B**

196. Which component serves as the standard, lightweight client-side load balancer in modern Spring Cloud architectures?

- A) Netflix Ribbon
- B) Spring Cloud LoadBalancer
- C) AWS ELB
- D) Nginx Router
- **Answer: B**

197. What is the benefit of a centralized API Gateway in a microservice mesh?

- A) It replaces the need for individual service codebases.
- B) It provides a single entry point to handle common concerns like cross-cutting security, routing, rate limiting, and metrics.
- C) It speeds up local compilation workflows.
- D) It enforces desktop-first user layout patterns.
- **Answer: B**

198. What distributed tracking concept groups related asynchronous network operations together under a single identifier?

- A) Span ID
- B) Trace ID
- C) Log Sequence Number
- D) Component Identifier
- **Answer: B**

199. What does an individual Span ID represent within a distributed trace graph?

- A) The total time an application takes to boot up.
- B) A single, discrete unit of work or network hop within a larger transaction path.
- C) The size of a database data record block.
- D) The number of lines of code in a class.
- **Answer: B**

200. What methodology defines twelve best practices for building scalable, independent, and maintainable cloud-native applications?

- A) Agile Manifesto
- B) The 12-Factor App methodology
- C) SOLID Design Principles
- D) Object-Oriented Design Rules

- **Answer: B**

Core Java Basics & Control Structures (Questions 201–210)

201. Which primitive data type in Java occupies exactly 2 bytes of memory and holds Unicode characters?

- A) byte
- B) short
- C) char
- D) int
- **Answer: C**

202. What happens when a Java loop attempts to access an array index that is equal to the length of the array?

- A) It reads a default value of zero.
- B) The loop terminates gracefully without outputting anything.
- C) A runtime `ArrayIndexOutOfBoundsException` is thrown.
- D) The array expands dynamically.
- **Answer: C**

203. Which control flow statement is ideal for executing alternative code blocks based on a matching scalar value or string?

- A) if...else
- B) switch
- C) while
- D) do...while
- **Answer: B**

204. What is the output of the expression $5 + 3 * 2$ in Java?

- A) 16
- B) 11
- C) 13
- D) 25
- **Answer: B**

205. Which operator is used to check if an object is an instance of a specific class or interface at runtime?

- A) typeof
- B) instanceof
- C) is
- D) extends
- **Answer: B**

206. What access modifier gives a class member the widest possible visibility across all packages?

- A) private
- B) protected
- C) Default (package-private)
- D) public
- **Answer: D**

207. Which keyword prevents a variable's value from being modified after it has been initialized?

- A) static
- B) final
- C) volatile
- D) abstract
- **Answer: B**

208. What is the default value of an uninitialized instance variable of an object reference type?

- A) ""
- B) 0
- C) null
- D) false
- **Answer: C**

209. Which memory area in the JVM is used to allocate space for objects and their instance variables?

- A) Stack Memory
- B) Heap Memory
- C) Method Area
- D) PC Register
- **Answer: B**

210. What type of variables are allocated on the JVM Stack?

- A) Instance variables
- B) Local variables and reference variables inside a method
- C) Static class variables
- D) Metadata definitions
- **Answer: B**

Object-Oriented Principles & Collection Internals (Questions 211–220)

211. What OOP principle focuses on hiding internal implementation details and exposing only necessary operations through an interface?

- A) Inheritance
- B) Polymorphism
- C) Abstraction
- D) Coupling
- **Answer: C**

212. What rule applies to overriding methods regarding access visibility?

- A) The overriding method can narrow the visibility of the superclass method.
- B) The overriding method must have the exact same visibility or broader visibility than the superclass method.
- C) The overriding method must be declared private.
- D) Visibility constraints do not apply to method overriding.
- **Answer: B**

213. Which class should be used to concatenate strings in a single-threaded loop where performance is critical?

- A) String
- B) StringBuffer
- C) StringBuilder
- D) StringJoiner
- **Answer: C**

214. Which method of the Object class should be overridden alongside equals() to ensure correct behavior in hash-based collections?

- A) toString()
- B) hashCode()
- C) clone()
- D) finalize()
- **Answer: B**

215. What is the underlying data structure of a standard Java ArrayList?

- A) Linked List
- B) Resizable Array
- C) Hash Table
- D) Red-Black Tree
- **Answer: B**

216. Which collection implementation guarantees that elements are maintained in a sorted order?

- A) HashSet
- B) LinkedHashSet
- C) TreeSet
- D) ArrayList
- **Answer: C**

217. What occurs if you modify a collection structurally while iterating over it using a standard fail-fast Iterator?

- A) The collection handles the modification seamlessly.
- B) A ConcurrentModificationException is thrown.
- C) The loop skips elements silently.
- D) The program hangs indefinitely.
- **Answer: B**

218. Which functional interface method does a Lambda expression map to?

- A) The single abstract method of the interface
- B) Any default method
- C) The constructor method
- D) All static methods
- **Answer: A**

219. What does the Stream collect(Collectors.toList()) operation represent?

- A) An intermediate operation
- B) A terminal operation that reduces stream data into a List container
- C) A parallel distribution filter
- D) A mapping transform operation
- **Answer: B**

220. Which wrapper class corresponds to the char primitive type?

- A) Char
- B) Character
- C) String
- D) Byte
- **Answer: B**

Enterprise Java: Web Components & Dependency Management (Questions 221–230)

221. What scope keeps a Spring bean instance alive for the duration of a single HTTP request?

- A) Singleton
- B) Prototype
- C) Request
- D) Session
- **Answer: C**

222. Where does Maven download and cache remote repository artifacts locally on your machine?

- A) The target folder
- B) The .m2 repository folder
- C) The src directory
- D) The workspace properties file
- **Answer: B**

223. What method initiates a Servlet instance when it is first loaded by the web container?

- A) service()
- B) init()
- C) doGet()

- D) destroy()
- **Answer: B**

224. Which implicit object in JSP represents the current page's servlet execution environment context output stream?

- A) request
- B) response
- C) out
- D) session
- **Answer: C**

225. What annotation marks a class as a Spring MVC controller that returns logical view names?

- A) @Component
- B) @Service
- C) @Controller
- D) @RestController
- **Answer: C**

226. Which annotation configures a method to handle HTTP GET requests for a specific URL path in Spring MVC?

- A) @PostMapping
- B) @GetMapping
- C) @PutMapping
- D) @DeleteMapping
- **Answer: B**

227. What annotation automatically generates setter and getter methods, constructors, and builders during compilation?

- A) @Autowired
- B) @Component
- C) @Data (from Project Lombok)
- D) @Entity
- **Answer: C**

228. What component acts as the persistence context for managing entities during an active transaction block in JPA/Hibernate?

- A) SessionFactory
- B) EntityManager (or Session)
- C) Connection
- D) Statement
- **Answer: B**

229. Which annotation maps a Java entity property to a specific column name in a database table?

- A) @Id
- B) @Table
- C) @Column
- D) @Transient
- **Answer: C**

230. What annotation prevents a class field from being persisted to the database by Hibernate?

- A) @Column
- B) @Transient
- C) @Id
- D) @Embedded
- **Answer: B**

Spring Data JPA, H2 & Advanced Security (Questions 231–240)

231. Which interface can you extend to get standard CRUD operations out-of-the-box without writing implementation code?

- A) CrudRepository or JpaRepository
- B) JdbcTemplate
- C) NamedParameterJdbcTemplate
- D) Session
- **Answer: A**

232. How can you define a custom database query method in a Spring Data JPA repository interface without writing SQL?

- A) By using a specific method naming convention that Spring Data parses into a query automatically.
- B) By writing documentation comments above the method.

- C) By declaring the method as static.
- D) Spring Data does not support custom queries without explicit SQL.
- **Answer: A**

233. What annotation allows you to define a manual JPQL or native SQL query directly above a repository method?

- A) @Param
- B) @Query
- C) @Modify
- D) @Select
- **Answer: B**

234. What type of database is an embedded MongoDB setup used for local integration tests?

- A) Relational SQL Database
- B) NoSQL Document Database
- C) Graph Database
- D) Key-Value Store
- **Answer: B**

235. What is the role of a Mock object when unit testing a Service class using Mockito?

- A) It connects to the production database to verify data.
- B) It replaces a real dependency (like a repository) with a controlled test double to isolate the class under test.
- C) It speeds up the local network routing.
- D) It minifies the test code files.
- **Answer: B**

236. Which annotation initializes a mock instance automatically when using Mockito in unit tests?

- A) @InjectMocks
- B) @Mock
- C) @Test
- D) @Autowired
- **Answer: B**

237. Which annotation injects defined mock instances into the target service instance being tested?

- A) @Mock
- B) @InjectMocks
- C) @Captor
- D) @Spy
- **Answer: B**

238. What does stateless authentication mean when using JSON Web Tokens (JWT)?

- A) The server stores token data in a local database session table.
- B) The server does not store session state; all necessary user identity data is embedded securely within the incoming token itself.
- C) Users must log in every time they click a button.
- D) Security credentials are passed as clear-text URL parameters.
- **Answer: B**

239. What are the three parts that make up a standard JSON Web Token (JWT), separated by dots?

- A) Name, Date, Key
- B) Header, Payload, Signature
- C) Username, Password, Role
- D) Issuer, Subject, Expiration
- **Answer: B**

240. Which Spring Security component intercepts incoming requests to extract tokens and verify user identity?

- A) AuthenticationManager
- B) A custom Security Filter (extending OncePerRequestFilter)
- C) SecurityContextHolder
- D) AuthenticationEntryPoint
- **Answer: B**

Cloud Native Operations, Eureka & Distributed Architecture (Questions 241–250)

241. What is the primary drawback of a monolithic application architecture as it scales?

- A) It is easy to deploy initially.
- B) Scaling the application requires scaling the entire monolith, and a single bug can bring down the whole system.
- C) It uses too few classes.
- D) It does not support relational databases.
- **Answer: B**

242. What happens to a microservice instance when it boots up in an environment managed by a Eureka Server?

- A) It downloads all database tables.
- B) It registers its network location and metadata with the Eureka registry.
- C) It shuts down all other running instances.
- D) It creates an external Git repository.
- **Answer: B**

243. How frequently do microservice instances send heartbeats to Eureka Server to confirm they are still healthy?

- A) Every 24 hours
- B) Periodically at configured intervals (default is 30 seconds)
- C) Only when receiving an external client request
- D) Never, registration happens only once at startup
- **Answer: B**

244. What happens if Eureka Server misses multiple heartbeats from a registered microservice instance?

- A) It alerts the cloud administrator's phone.
- B) It removes the unhealthy instance from its registry automatically.
- C) It restarts the entire cloud cluster.
- D) It updates the service's source code.
- **Answer: B**

245. What annotation enables an application to act as a Eureka Service Registry?

- A) `@EnableEurekaClient`

- B) @EnableEurekaServer
- C) @EnableDiscoveryClient
- D) @EnableGateway
- **Answer: B**

246. What annotation registers a microservice as a client in a Spring Cloud discovery registry?

- A) @EnableEurekaServer
- B) @EnableDiscoveryClient (or @EnableEurekaClient)
- C) @EnableFeignClients
- D) @EnableCircuitBreaker
- **Answer: B**

247. Which property sets the destination location backend repository for a Spring Cloud Configuration Server?

- A) spring.cloud.config.server.git.uri
- B) server.port
- C) spring.datasource.url
- D) eureka.client.service-url.defaultZone
- **Answer: A**

248. What is the role of a fallback method in a Resilience4J circuit breaker configuration?

- A) It retries the failed operation infinitely.
- B) It executes alternative logic immediately when the main service call fails or the circuit is open.
- C) It throws a runtime exception to crash the application.
- D) It formats the error message into an HTML table layout.
- **Answer: B**

249. Which HTTP method should be used to retrieve resource data without causing any side effects on the server?

- A) POST
- B) PUT
- C) GET
- D) DELETE
- **Answer: C**

250. Which HTTP method is designed to create a new resource at a server-determined URL destination path?

- A) GET
- B) POST
- C) PUT
- D) OPTIONS
- **Answer: B**

Client Interface Engineering & Layout Systems (Questions 251–260)

252. Which HTML5 tag defines a navigation block containing links to other pages or sections within a website?

- A) <section>
- B) <nav>
- C) <aside>
- D) <header>
- **Answer: B**

253. What CSS property determines the layering order of overlapping elements along the Z-axis?

- A) position
- B) z-index
- C) display
- D) float
- **Answer: B**

254. What values are acceptable for the CSS position property?

- A) static, relative, absolute, fixed, sticky
- B) inline, block, flex, grid
- C) top, bottom, left, right
- D) bold, italic, underline
- **Answer: A**

255. What target viewport metric does a responsive design media query typically evaluate?

- A) Server response speed

- B) Screen width or height (max-width, min-width)
- C) Browser history length
- D) Database row limits
- **Answer: B**

256. What unit represents a relative sizing scale based on the root element's font size in CSS?

- A) px
- B) em
- C) rem
- D) %
- **Answer: C**

257. Which Bootstrap class applies a flexible, responsive layout container that spans the full width of the viewport?

- A) .container
- B) .container-fluid
- C) .row
- D) .col-12
- **Answer: B**

258. How do you create a standard row layout structure in Bootstrap to wrap individual grid columns?

- A) Use the .row class.
- B) Use the .grid class.
- C) Use the .container class.
- D) Use the .navbar class.
- **Answer: A**

259. What utility class in Bootstrap sets text alignment to the right or end of an element?

- A) .text-start
- B) .text-center
- C) .text-end
- D) .text-justify
- **Answer: C**

260. Which Bootstrap button class renders a clear, standard primary call-to-action button?

- A) .btn-secondary
- B) .btn-success
- C) .btn-primary
- D) .btn-danger
- **Answer: C**

JavaScript Event Logic & Functional Mechanisms (Questions 261–270)

261. What does the JavaScript method `document.getElementById()` return?

- A) An array of matching objects.
- B) A single DOM element matching the specified ID attribute value.
- C) A JSON data string.
- D) A Promise object.
- **Answer: B**

262. Which method adds an event listener function to a DOM element without overriding existing event handlers?

- A) `onclick`
- B) `addEventListener()`
- C) `attachEvent()`
- D) `dispatchEvent()`
- **Answer: B**

263. What method stops an event from bubbling up the DOM tree to parent elements?

- A) `event.preventDefault()`
- B) `event.stopPropagation()`
- C) `event.stopImmediatePropagation()`
- D) `return false;`
- **Answer: B**

264. What method cancels the default browser behavior associated with an event, such as a form submission reloading the page?

- A) `event.stopPropagation()`

- B) event.preventDefault()
- C) event.cancelBubble()
- D) event.halt()
- **Answer: B**

265. What is the scope of a variable declared with the var keyword inside a function block?

- A) Global scope
- B) Function scope
- C) Block scope
- D) Module scope
- **Answer: B**

266. What does the strict equality operator === evaluate in JavaScript?

- A) It compares values for equality after performing type conversion.
- B) It compares both the value and the data type for equality without type conversion.
- C) It checks if variables are declared constants.
- D) It assigns a value to a variable.
- **Answer: B**

267. Which method converts a native JavaScript object into a JSON-formatted text string?

- A) JSON.parse()
- B) JSON.stringify()
- C) JSON.toString()
- D) JSON.format()
- **Answer: B**

268. What keyword references the execution context object of a running JavaScript function?

- A) this
- B) super
- C) parent
- D) window
- **Answer: A**

269. Which method executes a callback function once after a specified delay in milliseconds?

- A) setInterval()
- B) setTimeout()
- C) clearTimeout()
- D) delay()
- **Answer: B**

270. Which method cancels a repeating timer operation that was established using setInterval()?

- A) clearTimeout()
- B) clearInterval()
- C) stopInterval()
- D) pause()
- **Answer: B**

Enterprise REST Architectures & Integration Testing (Questions 271–280)

271. Which HTTP method is designed to update an existing resource completely or create it if it does not exist?

- A) GET
- B) POST
- C) PUT
- D) DELETE
- **Answer: C**

272. Which HTTP status code indicates that a request completed successfully?

- A) 200 OK
- B) 201 Created
- C) 400 Bad Request
- D) 500 Internal Server Error
- **Answer: A**

273. Which HTTP status code indicates that a client request is missing required authentication credentials?

- A) 400 Bad Request

- B) 401 Unauthorized
- C) 403 Forbidden
- D) 404 Not Found
- **Answer: B**

274. Which HTTP status code indicates that a server cannot find the requested resource URL path?

- A) 200 OK
- B) 400 Bad Request
- C) 404 Not Found
- D) 500 Internal Server Error
- **Answer: C**

275. Which HTTP status code indicates a general, unhandled runtime error on the server?

- A) 400 Bad Request
- B) 401 Unauthorized
- C) 403 Forbidden
- D) 500 Internal Server Error
- **Answer: D**

276. Which annotation slices out the HTTP response status code to return from a controller method explicitly?

- A) @ResponseStatus
- B) @ResponseBody
- C) @RequestMapping
- D) @ExceptionHandler
- **Answer: A**

277. What framework annotation bootstraps an integration test context for a Spring Boot application?

- A) @SpringBootTest
- B) @WebMvcTest
- C) @DataJpaTest
- D) @ExtendWith
- **Answer: A**

278. Which annotation tests only the web layer components (controllers) while ignoring database components to keep tests fast?

- A) @SpringBootTest
- B) @WebMvcTest
- C) @DataJpaTest
- D) @RestClientTest
- **Answer: B**

279. Which utility class acts as a mock entry point to perform fake HTTP requests and verify controller responses without running a full server?

- A) RestTemplate
- B) MockMvc
- C) WebTestClient
- D) HttpClient
- **Answer: B**

280. What annotation auto-configures the MockMvc instance within a Spring test class structure?

- A) @SpringBootTest
- B) @AutoConfigureMockMvc
- C) @Mock
- D) @InjectMocks
- **Answer: B**

Cloud Routing, Gateways & Enterprise Architecture (Questions 281–290)

281. What architectural style separates data modifications (mutations) from data reads (queries) into distinct models?

- A) Monolithic Pattern
- B) CQRS (Command Query Responsibility Segregation)
- C) MVC Pattern
- D) Facade Pattern
- **Answer: B**

282. What does an API Gateway use to decide which backend microservice should handle an incoming request path?

- A) Circuit Breaker thresholds
- B) Route Predicates and Filters
- C) Database connection pools
- D) Maven build targets
- **Answer: B**

283. What component can modify an incoming HTTP request header or outgoing response header within an API Gateway?

- A) Service Registry
- B) Gateway Filter
- C) Circuit Breaker
- D) Dependency Configurator
- **Answer: B**

284. Which annotation enables Spring Cloud Configuration client discovery updates inside a calling service?

- A) @EnableConfigServer
- B) @SpringBootApplication
- C) No explicit annotation is needed; adding the starter client dependency resolves it automatically via auto-configuration.
- D) @RefreshScope
- **Answer: C**

285. What pattern coordinates transactions across multiple independent microservices to maintain data consistency?

- A) Saga Pattern
- B) Singleton Pattern
- C) Factory Pattern
- D) Observer Pattern
- **Answer: A**

286. What is the default port used by a Spring Cloud Netflix Eureka Server application?

- A) 8080
- B) 8761
- C) 9090
- D) 3306

- **Answer: B**

287. What is the default port used by a Spring Cloud Configuration Server instance?

- A) 8761
- B) 8888
- C) 8080
- D) 1521
- **Answer: B**

288. Which annotation marks an interface as a declarative REST client using OpenFeign?

- A) @RestController
- B) @FeignClient
- C) @Autowired
- D) @Service
- **Answer: B**

289. What property matches a Feign Client interface to its target service registration ID in Eureka?

- A) value or name matching the service application ID
- B) url matching a hardcoded database location
- C) path pointing to local test folders
- D) scope setting it to prototype
- **Answer: A**

290. What strategy distributes incoming network traffic uniformly across a pool of healthy backend service instances?

- A) Circuit Breaking
- B) Distributed Tracing
- C) Load Balancing
- D) Autowiring
- **Answer: C**

Comprehensive Mastery Review (Questions 291–300)

291. What happens during the Compilation phase of a Java application?

- A) Source code .java files are compiled into platform-independent .class bytecode files.
- B) Bytecode is executed directly inside operating system kernels.
- C) Objects are allocated space inside JVM Heap areas.
- D) Database schemas are generated automatically.
- **Answer: A**

292. Which keyword terminates the execution of a surrounding loop or switch block immediately?

- A) continue
- B) break
- C) return
- D) exit
- **Answer: B**

293. Which keyword skips the remaining statements in the current iteration of a loop and proceeds to the next evaluation cycle?

- A) break
- B) continue
- C) pass
- D) skip
- **Answer: B**

294. What does a non-zero exit status code signify when a command-line script or build process completes?

- A) The process completed successfully without any issues.
- B) An error or abnormal termination occurred during execution.
- C) The process requires more memory resources.
- D) The operating system completed a background update.
- **Answer: B**

295. Which type of database relationship maps a user record to a single unique profile record?

- A) One-to-Many (@OneToMany)
- B) Many-to-One (@ManyToOne)
- C) One-to-One (@OneToOne)
- D) Many-to-Many (@ManyToMany)

- **Answer: C**

296. What is the purpose of the CSS `box-sizing: border-box;` property declaration?

- A) It completely removes margins from elements.
- B) It includes padding and borders within an element's total calculated width and height, preventing sizing issues.
- C) It changes block elements into inline layouts.
- D) It applies a default outline border to elements.
- **Answer: B**

297. Which HTML5 semantic tag designates footer content, such as copyright notices or contact info, at the bottom of a page or section?

- A) `<header>`
- B) `<section>`
- C) `<footer>`
- D) `<bottom>`
- **Answer: C**

298. Which method converts a JSON-formatted string into a native JavaScript object structure?

- A) `JSON.stringify()`
- B) `JSON.parse()`
- C) `Object.parse()`
- D) `String.toJSON()`
- **Answer: B**

299. What approach focuses on writing test cases for a piece of functionality before writing the actual implementation code?

- A) Behavior Driven Development (BDD)
- B) Test-Driven Development (TDD)
- C) Integration Testing
- D) Manual regression scanning
- **Answer: B**

300. What does the acronym REST stand for in web service design patterns?

- A) Rational Entry System Transfer
- B) Representational State Transfer
- C) Remote Environment Storage Transaction
- D) Realtime Encrypted Stream Transmission
- **Answer: B**